Skew-Aware Federated Learning: Addressing Lesion-Size Heterogeneity in Medical Image Segmentation

Source code: https://github.com/sameelhaider32/lesion-skewed-fl-segmentation

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Abstract

Hospitals cannot share patient MRI data with each other due to privacy laws, but they still need to train good brain tumour segmentation models. Federated learning solves this by letting hospitals share model weights instead of raw data. The problem is that standard federated learning treats all hospitals equally at aggregation: it weights each one only by how many cases they have. This is unfair for small tumours ($<30\,\mathrm{cm}^3$), which are rare and concentrated at only a few hospitals. Those hospitals end up with very little influence on the final model, so the model never learns to segment small tumours well.

We propose SCWA-LW (Size-Composition Weighted Aggregation with Layer-Wise routing). The idea is simple: give hospitals with more small-tumour cases a bigger say when updating the decoder of the segmentation network, while keeping encoder aggregation as standard FedAvg. The encoder learns general brain anatomy that all hospitals contribute to equally. The decoder draws the actual tumour boundary, so it should hear more from the hospitals that have actually seen small tumours.

We ran four experiments to find out what actually drives improvement. A stronger loss function alone barely helped. Adding FedProx regularisation without SCWA-LW actually made things worse; it stopped the small-tumour-rich hospital from diverging in a way that was actually useful. Adding SCWA-LW on top was what made the difference: $+0.021$ small-tumour Dice and $+0.036$ fair score over the no-SCWA control. Our full system achieves a 10% relative improvement in small-tumour Dice and 8.3% in fair score over the FedAvg baseline on the FeTS 2022 benchmark. Each hospital only needs to share one number with the server (how many small-tumour cases they have), so the privacy cost is essentially zero.

1. Introduction

Brain tumour segmentation from multi-parametric MRI is a clinically critical task, providing the volumetric measurements used to guide surgical planning, radiotherapy target delineation, and treatment response monitoring. Deep learning models trained on large, diverse datasets have demonstrated strong performance on this task (Isensee et al., 2021), but in practice, patient data cannot be pooled freely across hospital sites due to privacy regulations and data governance constraints. Federated learning (FL) addresses this by training a shared model through iterative exchange of model weights rather than raw data (McMahan et al., 2017), making it a natural fit for multi-site medical image analysis.

A well-known challenge in federated learning is that hospitals have very different patient populations, so the data at each site is non-IID (Li et al., 2020). In brain tumour segmentation this creates a specific problem: small tumours ($<30\,\mathrm{cm}^3$) are rare and tend to be concentrated at just one or two hospitals. Under standard Federated Averaging (McMahan et al., 2017), every hospital's weight in the aggregation is just proportional to how many cases it has. So a hospital with 20 small-tumour cases gets exactly the same vote as a hospital with zero. The result is a *fairness gap*: the global model learns to segment large tumours reasonably well but consistently fails on small ones.

Existing solutions try to fix this either by stabilising training across clients or by handling class imbalance within each client's local data. Neither approach fixes the real problem. Even if a client trains a great local model on small tumours, FedAvg still gives it the same aggregation weight as a client that has never seen one. The good local signal just gets averaged out. The fairness gap stays even when local training is improved.

Our fix is based on how the two halves of a U-Net work differently. The encoder (first half) learns general features about what brain tissue looks like. Every hospital contributes useful information here regardless of tumour size, so standard FedAvg makes sense for the encoder. The decoder (second half) is what actually draws the tumour boundary. Segmenting a tiny $3\,\mathrm{cm}^3$ lesion versus a large $200\,\mathrm{cm}^3$ mass requires very different boundary reasoning, so the decoder benefits much more from hospitals that have actually seen small tumours. This is exactly where the aggregation bias hurts.

We propose SCWA-LW: the encoder is aggregated with standard FedAvg every round, but the decoder is aggregated using a size-composition score that gives more weight to hospitals with more small-tumour cases. We combine this with a Focal Tversky loss to better supervise small lesions locally and FedProx to limit client drift. Together these form our complete system for fixing the small-tumour fairness gap in federated brain tumour segmentation.

Contributions. This paper makes the following contributions:

- We identify and quantify the small-lesion fairness gap in federated brain tumour segmentation under Dirichlet non-IID partitioning (α=0.5) and show that strengthening the local loss function alone is insufficient to close it.

- We propose SCWA-LW, a layer-wise aggregation strategy that routes decoder aggregation through size-composition-weighted client contributions, motivated by the distinct roles of encoder and decoder layers in segmentation networks.

- We demonstrate through a four-experiment controlled ablation on FeTS 2022 that SCWA-LW is the primary driver of improvement. A dedicated no-SCWA control (Exp B2: FedProx + Strong FTL, no SCWA-LW) performs *worse* than the FedAvg baseline on small tumours, confirming that SCWA-LW (not FedProx regularisation) accounts for the full $+0.021$ small-Dice and $+0.036$ fair-score gain.

- SCWA-LW requires no additional communication rounds, no per-round metadata, and only one integer per client at initialisation, making it compatible with standard FL privacy requirements.

2. Related Work

2.1. Brain Tumor Segmentation

Brain tumor segmentation in multi-modal MRI requires delineating sub-regions such as the whole tumor, tumor core, and enhancing tumor. The task is challenging due to extreme foreground-background imbalance and high variability in tumor size and shape across patients. Accurate delineation typically requires data from multiple institutions, motivating the use of federated learning in this domain.

2.2. Federated Learning for Medical Image Segmentation

Federated learning (FL) enables collaborative model training across hospitals without sharing raw patient data, addressing privacy regulations such as HIPAA and GDPR (Sun et al., 2025). FedAvg (McMahan et al., 2017) is the standard baseline, aggregating locally trained weights into a global model. Recent work such as FL-MedSegBench (Zhu et al., 2026) and FUSION (Raheem et al., 2025) shows that FL is viable for multi-institutional segmentation, but no single method dominates across all settings.

2.3. Hospital Heterogeneity and Client Drift

A core challenge in medical FL is that hospital data are non-IID: they differ in scanner type, imaging protocol, and patient population (Sun et al., 2025). This heterogeneity causes client drift, where local models diverge from the global objective during training. Regularization-based approaches such as FedProx address drift by penalizing local deviation from the global model, while SCAFFOLD corrects for gradient variance. Personalized methods, especially FedBN, which keeps batch normalization layers local, consistently outperform plain FedAvg in multi-hospital benchmarks (Zhu et al., 2026). Despite these advances, size bias persists: methods that stabilize training do not guarantee that small or rare lesions, typically concentrated in low-resource hospitals, receive adequate learning signal.

2.4. Domain Generalization to Unseen Hospitals

Strong in-federation performance does not imply generalization to new hospitals outside the training coalition. FedDG (Liu et al., 2021) addresses this via episodic learning in frequency space, improving robustness to domain shift on unseen sites. Similarly, Arasteh (Arasteh et al., 2023) show that federation diversity is the main driver of off-domain performance in chest radiograph analysis. In our setting, hospital-level lesion-size skew introduces an additional feature shift that further complicates generalization to new clinics.

2.5. Class and Client Imbalance in Federated Learning

Imbalance in FL takes multiple forms: local imbalance within a client, global imbalance across the federation, and mismatch between local and global distributions (Zhang et al., 2023). Wang (Wang et al., 2021) introduce Ratio Loss to handle minority classes and show that local and global imbalance require separate treatment. CLIMB (Shen et al., 2022) protects under-served clients through client-level loss constraints without requiring explicit minority identification. FedRS (Li & Zhan, 2021) addresses missing-class clients by restricting softmax updates for absent labels. FedIIC (Wu et al., 2023) further shows that imbalance degrades not only the classifier but also the quality of learned representations, which is especially damaging in medical imaging. These findings motivate our concern that lesion-size skew may cause analogous under-representation of small, hard-to-segment lesions in the federated segmentation setting.

2.6. Fairness and Worst-Client Performance

Average Dice score is a misleading metric when hospital distributions differ: a model can achieve high aggregate performance while failing on weak or data-scarce clients, which is a serious safety concern in clinical deployment. FL-MedSegBench (Zhu et al., 2026) explicitly reports worst-client Dice and client-wise variance alongside average metrics, highlighting that fairness across hospitals remains an open problem in federated segmentation.

2.7. Research Gap

Existing work addresses hospital heterogeneity, personalization, and client imbalance as separate concerns. However, no prior work jointly studies *lesion-size skew* and *quantity skew* as co-occurring client-level phenomena in federated segmentation. A federated model can still report acceptable average Dice while systematically failing on hospitals that hold small or emerging lesions and contribute fewer training cases. This gap motivates our work: we simulate this combined skew on the FeTS 2022 benchmark and evaluate whether a skew-aware training strategy can improve worst-client performance and small-lesion detection.

2.8. Research Question

The reviewed literature leaves a clear open problem: federated medical segmentation methods do not address lesion-size skew at the client level, and average Dice alone cannot reveal whether weak or data-scarce hospitals are being systematically underserved. We therefore ask:

> *Can a lesion-size-aware federated training strategy improve worst-client performance and small-lesion detection under hospital-level skew in medical image segmentation?*

3. Methodology

3.1. Dataset

We use the **FeTS 2022** federated brain tumor segmentation benchmark (Pati et al., 2022), which provides multi-modal MRI scans (FLAIR, T1, T1ce, T2) with voxel-level tumor masks from multiple institutions. Cases are grouped by lesion volume: *small* ($<$ 30 cm^3), *medium* (30–100 cm^3), and *large* ($>$ 100 cm^3). We simulate a 5-client federation with a locked test set shared across all experiments.

3.2. Data Partitioning

We partition the training pool across 5 clients using a **Dirichlet distribution** (α=0.5), applied independently per size group to induce non-IID skew in both lesion composition and client size. Each client is capped at 80 cases via stratified sampling. A stratified validation split of 10 cases per client is held out, guaranteeing small-lesion representation in every client's validation set.

Tumor cases are stratified into three size groups by whole-tumour volume, defined as an experimental design choice to partition the FeTS 2022 volume distribution into interpretable difficulty tiers (Pati et al., 2022):

- **Small:** $V < 30$ cm^3. Low-burden lesions with limited foreground voxels; hardest to segment and most affected by class imbalance.
- **Medium:** $30 \leq V < 100$ cm^3. Intermediate volume; the most frequent stratum in the FeTS 2022 training pool.
- **Large:** $V \geq 100$ cm^3. High-volume tumours; easiest to detect but clinically critical for surgical planning.

The resulting distribution (Table 1) shows strong non-IID skew: Client 0 is large-dominant (84% large), Clients 2–4 are medium-dominant, and Client 1 is the only client with substantial small-lesion representation (29% small), making it the primary source of small-lesion signal in the federation.

3.3. Training Configuration

All four experiments use exactly the same hyperparameters. This is important because if we changed things like the learning rate or number of rounds between experiments, we would not know whether any improvement came from our method or just from the different training setup. Each round, the server sends the current global model to all five clients. Each client trains on its own data and sends the updated weights back. The server then aggregates those weights into a new global model. Table 2 lists the full configuration.

3.4. Loss Functions

Small tumours occupy very few voxels compared to the overall scan. Because of this, standard Dice loss barely penalises the model for missing them entirely. If the model produces no prediction on a 3 cm^3 lesion, the Dice score barely moves because there were so few foreground voxels to begin with. The model learns it is "safe" to ignore small tumours. To fix this, we add a Focal Tversky Loss (FTL) (**?**), which controls the trade-off between false positives (predicting tumour where there is none) and false negatives (missing real tumour). We set it to penalise false negatives much more heavily, which forces the model to stop ignoring small lesions. The Tversky index is

Table 1. Client data distribution after Dirichlet (α=0.5) partitioning. Train/Val = number of cases. S/M/L = small/medium/large lesion counts within each split.

	Train				**Val**			
Client	**Total**	**S**	**M**	**L**	**Total**	**S**	**M**	**L**
Client 0	70	7	4	59	10	1	1	8
Client 1	70	20	26	24	10	3	4	3
Client 2	58	1	55	2	10	0	9	1
Client 3	70	9	46	15	10	1	7	2
Client 4	70	0	66	4	10	0	9	1
Total	**338**	**37**	**197**	**104**	**50**	**5**	**30**	**15**

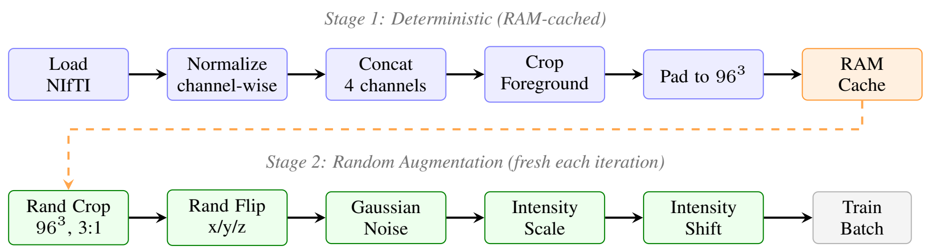

Figure 1. Two-stage MONAI preprocessing pipeline. Stage 1 is deterministic and cached in RAM; Stage 2 applies fresh random augmentation on every training iteration. Evaluation uses Stage 1 transforms only.

Table 2. Shared hyperparameters across all experiments.

Parameter	**Value**
Communication rounds	30 (Baseline: 50)
Local epochs per round	3
Optimiser	Adam
Learning rate	1×10^{-4}
Weight decay	1×10^{-5}
Batch size	2 (4 on A100)
Patches per case	4
Patch size	$96 \times 96 \times 96$
Sliding-window overlap (val)	0.50
Sliding-window overlap (test)	0.75
Model selection	Best val fair score

$$T(\hat{y}, y) = \frac{TP + \epsilon}{TP + \alpha\, FP + \beta\, FN + \epsilon}, \tag{1}$$

where $\alpha+\beta$=1 control the FP/FN trade-off. The Focal Tversky Loss applies a focal exponent γ to concentrate gradient on hard, under-segmented examples:

$$\mathcal{L}_{FTL} = \big(1 - T(\hat{y}, y)\big)^{\gamma}. \tag{2}$$

The combined per-client objective is

$$\mathcal{L} = \mathcal{L}_{Dice} + \lambda\, \mathcal{L}_{FTL}. \tag{3}$$

The baseline uses a mild setting (α=0.3, β=0.8, γ=2.0, λ=0.5) where the FTL contributes only a small amount to the total loss. Experiments B1, B2, and C use a *strong* configuration (α=0.3, β=0.85, γ=2.0, λ=2.0). This quadruples the FTL weight and increases the false-negative penalty, making the model pay much more attention to small lesions it would otherwise miss.

3.5. Federated Aggregation Strategies

FedAvg. Standard Federated Averaging (McMahan et al., 2017) weights each client proportionally to its dataset size:

$$w^{t+1} = \sum_{i=1}^{K} \frac{n_i}{N}\, w_i^t, \tag{4}$$

where n_i is client i's case count and N=$\sum_i n_i$. Because all clients have similar totals ($n_i \approx 70$), FedAvg weights are nearly uniform and ignore lesion-size composition.

FedProx. FedProx (Li et al., 2020) appends a proximal term to each client's local objective to limit divergence from the global model under non-IID data:

$$\mathcal{L}_i^{prox} = \mathcal{L}_i + \mu 2 \|w_i - w^t\|^2, \quad \mu\text{=}0.01. \tag{5}$$

SCWA-LW (Proposed). We propose *Size-Composition Weighted Aggregation with Layer-Wise routing* (SCWA-LW). The core idea is simple: not all clients are equally useful for learning to segment small tumours, so why treat them equally at aggregation? We give each client a weight that depends on both how many total cases it has *and* how many of those are small-tumour cases, blended equally via α=0.5:

$$\tilde{w}_i = (1-\alpha)\frac{n_i}{N} + \alpha\frac{s_i}{S}, \quad \alpha=0.5, \tag{6}$$

where s_i is client i's small-tumour case count and $S = \sum_i s_i$. As shown in Figure 2, Client 1 (20 small cases out of 70) goes from a 20.7% FedAvg weight up to 37.4%, while Client 4 (zero small cases) drops from 20.7% to 10.4%.

Crucially, these weights are *only applied to the decoder layers*. The encoder of the U-Net learns general features about brain anatomy and what tissue looks like. That kind of knowledge is useful from every client regardless of tumour size, so we leave encoder aggregation as standard FedAvg. The decoder, on the other hand, is responsible for actually drawing the segmentation boundary. Small and large tumours require very different boundary reasoning, so the decoder benefits directly from seeing more small-lesion examples. By routing the small-lesion-biased weights only through decoder aggregation, we boost small-tumour performance without disrupting the shared representations that all clients contribute to.

The weights are computed *once* before training starts. Each client only needs to share one number with the server: s_i, their small-tumour case count. No extra communication is needed during training rounds.

3.6. Evaluation Metrics

We report four complementary metrics computed on a locked 90-case test set (30 small / 30 medium / 30 large), evaluated with sliding-window inference at overlap 0.75:

- **Per-group Dice** ($Dice_{small}$, $Dice_{medium}$, $Dice_{large}$). The standard volumetric overlap metric broken down by tumour-size group. Reporting per-group scores reveals whether a method genuinely improves small-lesion performance or merely redistributes gains from larger tumours.
- **Fair score.** The harmonic mean of the three per-group Dice values: $\mathcal{F} = 3 / (1/D_S + 1/D_M + 1/D_L)$. The harmonic mean penalises imbalance more severely than the arithmetic mean: a model that scores well on large tumours but poorly on small ones receives a low fair score even if its overall mean is high. We use $\mathcal{F}$ for model selection during validation.
- **Overall mean Dice.** The case-level mean across all 90 test cases regardless of group, reflecting general segmentation quality.
- **Fairness gap.** $\Delta = 12(D_M + D_L) - D_S$. A positive gap indicates how much the small-tumour group lags behind the others. Reducing Δ is the direct objective of this work.

Table 3. Ablation design. Each row adds one component over the previous, enabling isolated attribution.

Experiment	Strong FTL	FedProx	SCWA-LW
A: FedAvg Baseline	×	×	×
B1: Strong FTL		×	×
B2: FedProx + FTL			×
C: Full System			

3.7. Experimental Design

We designed four experiments to isolate exactly which components of our system actually help (Table 3).

Experiment A is the FedAvg baseline with a standard mild loss. It gives us a reference number to measure all improvements against.

Experiment B1 keeps standard FedAvg aggregation but switches to the stronger loss. The question this answers is: can we close the fairness gap just by making each client train better locally, without changing aggregation at all?

Experiment B2 adds FedProx on top of B1's loss but still without SCWA-LW. This is a critical control experiment. FedProx is a well-known technique for federated learning with non-IID data, so it is important to know whether it helps or hurts in our setting before adding SCWA-LW.

Experiment C is the full system: same as B2 but with SCWA-LW applied to decoder aggregation. Because B2 and C are identical in everything except the aggregation weights, comparing them tells us exactly how much SCWA-LW contributes on its own.

The test set and random seed are the same across all four experiments so results are directly comparable.

4. Results

4.1. Quantitative Results

Table 4 reports test-set performance for all four experiments alongside a FedProx-only baseline from a prior phase of this project, which uses a smaller model (channels 16–256, ≈4.7M parameters), standard Dice loss, and one local epoch per round. Our full system (Exp C) outperforms this prior baseline on every metric, with a 45% relative improvement in small-tumour Dice (from 0.088 to 0.128).

Figure 3 visualises the same results as a grouped bar chart, making the per-group gain pattern across experiments easier to compare at a glance.

4.2. Ablation Analysis

Table 5 decomposes gains into individual component contributions. The four-experiment design allows each factor to

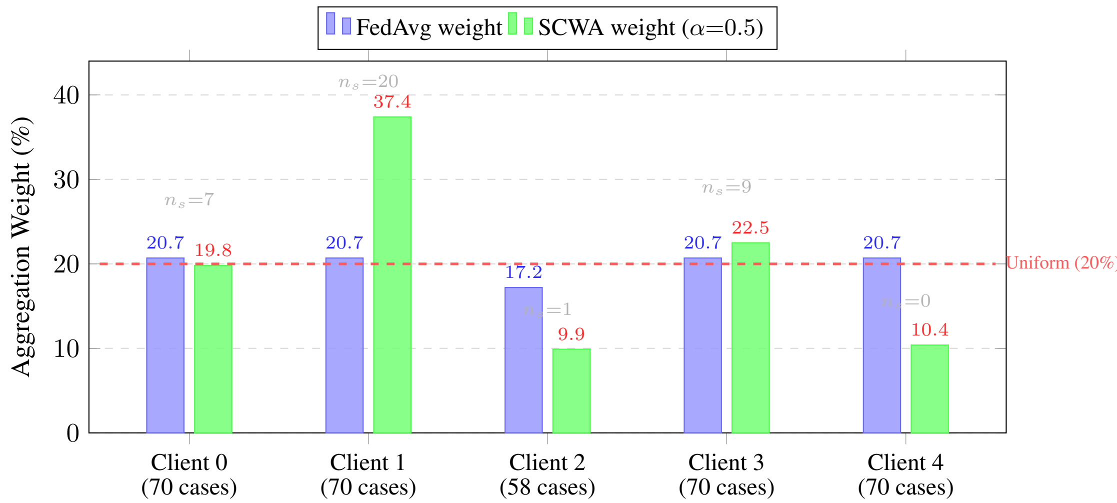

Figure 2. Aggregation weight per client under standard FedAvg (blue) and SCWA with α=0.5 (green). n_s denotes the number of small-tumour training cases per client. Client 1, the only client with substantial small-lesion data (20 of 70 cases), gains 16.7 percentage points of weight. Clients 2 and 4, with one and zero small cases respectively, are downweighted relative to FedAvg. The dashed line marks the uniform reference weight (20%).

Table 4. Test-set Dice scores. Best values among our experiments in **bold**. Fair = harmonic mean(S,M,L). Gap = $12(D_M{+}D_L) - D_S$.

Method	**Small**	**Med.**	**Large**	**Fair**	**Gap**
Prior baseline (FedProx)	0.088	0.326	0.599	0.190	
A: FedAvg Baseline	0.116	0.413	0.694	0.241	0.437
B1: Strong FTL	0.118	0.446	0.704	0.247	0.457
B2: FedProx + FTL	0.107	0.402	0.681	0.225	0.435
C: Full System	**0.128**	0.435	**0.716**	**0.261**	0.448
C vs A	*+0.012*	*+0.022*	*+0.022*	*+0.020*	—
C vs B2 (pure SCWA)	*+0.021*	*+0.033*	*+0.035*	*+0.036*	

Table 5. Isolated contribution of each component. Δ values are signed differences in test Dice.

Component	**ΔSmall**	**ΔFair**	**ΔMean**
Strong FTL only (B1 vs. A)	+0.0017	+0.0065	+0.0150
FedProx w/o SCWA (B2 vs. A)	−0.0095	−0.0162	−0.0113
Pure SCWA-LW (C vs. B2)	+0.0215	+0.0358	+0.0298
Full system (C vs. A)	+0.0115	+0.0198	+0.0185

be isolated cleanly.

Strong FTL alone (B1 vs. A). Adding the strong Focal Tversky loss yields only marginal improvement: +0.0017 in small-Dice and +0.0065 in fair score. While helpful, stronger local supervision alone is far from sufficient to bridge the small-lesion fairness gap.

FedProx without SCWA (B2 vs. A). Adding FedProx regularisation (without any change to aggregation) actually *hurts*: small-Dice drops by 0.009 and fair score drops by 0.016 relative to the FedAvg baseline. FedProx penalises every client for diverging from the global model, including Client 1 whose divergence is beneficial because it holds the most small-tumour cases. Suppressing that divergence removes the very signal we need.

Pure SCWA-LW contribution (C vs. B2). This is the cleanest test: B2 and C share identical loss and regularisation. The only difference is whether SCWA-LW is applied at aggregation. Adding SCWA-LW delivers +0.021 in small-Dice and +0.036 in fair score. This is the dominant contribution and confirms that the aggregation strategy (not any local training change) is the primary driver of small-tumour performance.

Full system (C vs. A). The complete system achieves a 10% relative improvement in small-tumour Dice and an 8.3% improvement in fair score over the FedAvg baseline.

4.3. Training Dynamics

All four methods converge by round 20, which is consistently the best-checkpoint round across experiments. Experiment C reaches the highest validation fair score at every evaluation point beyond round 10. The small-tumour Dice in Experiment C at round 20 (val=0.104, test=0.128) exceeds both the baseline and Experiment B1 test values, confirming that the gain is not an artefact of the evaluation schedule. A slight regression at round 30 is consistent with mild late-round overfitting and justifies the harmonic-mean

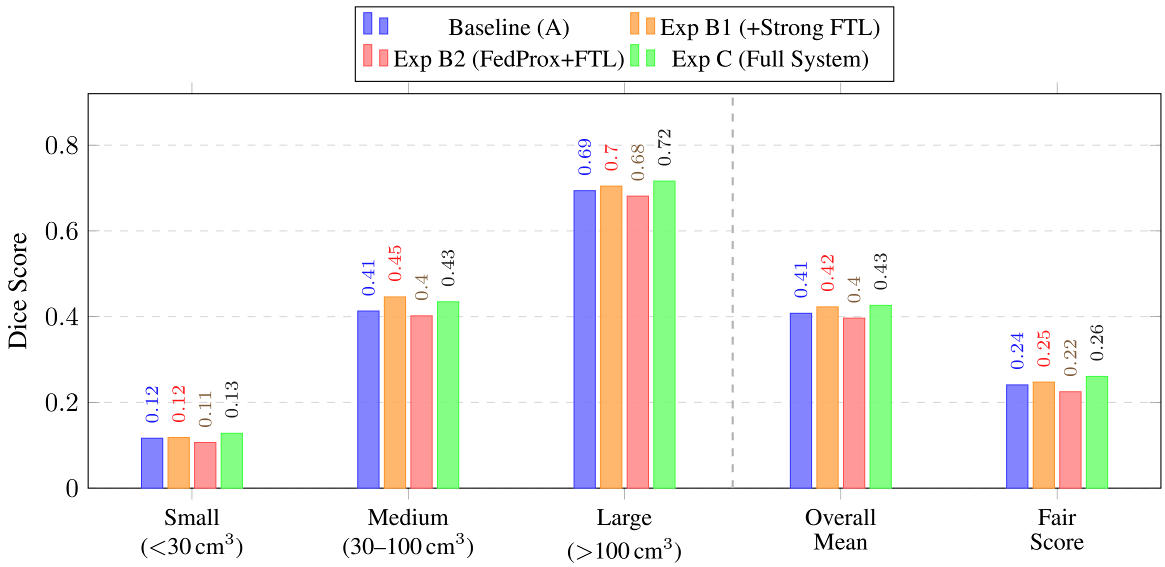

Figure 3. Test-set Dice scores by tumour size group and summary metrics for all four experiments. The dashed vertical line separates per-group scores (Small, Medium, Large) from aggregate metrics (Overall Mean, Fair Score). Experiment B2 (FedProx without SCWA-LW) scores *below* Baseline A on every metric, confirming that FedProx alone is harmful. Experiment C achieves the highest Small, Large, and Fair scores; Experiment B1 leads on Medium Dice owing to the stronger FTL weight alone.

model selection strategy.

4.4. Qualitative Results

On a representative small-tumour test case, the baseline model produces no foreground at all; it misses the lesion entirely. Experiment C localises the same lesion with visible, if imperfect, overlap. On medium-tumour cases both models produce reasonable masks, with Experiment C showing tighter boundary adherence. This qualitative pattern is consistent with the quantitative small-tumour Dice gap (0.116 vs. 0.128) reported in Table 4.

5. Discussion

Why SCWA-LW drives the improvement. The four-experiment ablation tells a very clear story. Experiment B1 (strong FTL only) barely moved small-tumour Dice: just +0.002 over the baseline. More striking, Experiment B2 added FedProx on top of B1 but left out SCWA-LW, and it actually performed *worse* than the FedAvg baseline on small tumours (0.107 vs. 0.116). Why? FedProx penalises Client 1 for diverging from the global model, but Client 1's divergence is exactly what we want because it holds the most small-tumour cases. Suppressing that useful divergence without fixing the aggregation bias just makes things worse. The real improvement came from adding SCWA-LW in Experiment C. Comparing C to B2 gives us the cleanest possible test because the only difference between the two experiments is the aggregation weights. That single change delivers +0.021 small-Dice and +0.036 fair score. SCWA-LW gives Client 1 a much bigger say in decoder aggregation, so the small-tumour signal that FedProx was holding back actually gets used to update the model.

Why the fairness gap does not fully close. Even though small-tumour Dice improved, the fairness gap did not disappear. Experiment B1 widened the gap from 0.437 to 0.457 because medium and large tumour performance improved faster than small. Experiment B2 brought it back to 0.435 (nearly identical to Baseline A), but only by dragging *all* metrics down together. Experiment C lands at 0.448 while simultaneously achieving the best small-tumour Dice (0.128), meaning the remaining gap reflects genuine medium and large gains rather than small-tumour failure. The honest reason the gap persists is data scarcity: only 37 out of 338 training cases are small-tumour (11%). That is a hard ceiling. No aggregation strategy can manufacture signal that is not in the data.

Privacy. SCWA adds almost no privacy cost. Each client shares one integer with the server at setup: how many small-tumour cases they have. Nothing extra is exchanged during training rounds beyond the standard weight updates that FedAvg already requires.

Limitations

- **Static weights.** The aggregation weights are computed once before training starts and never updated. If a hospi-

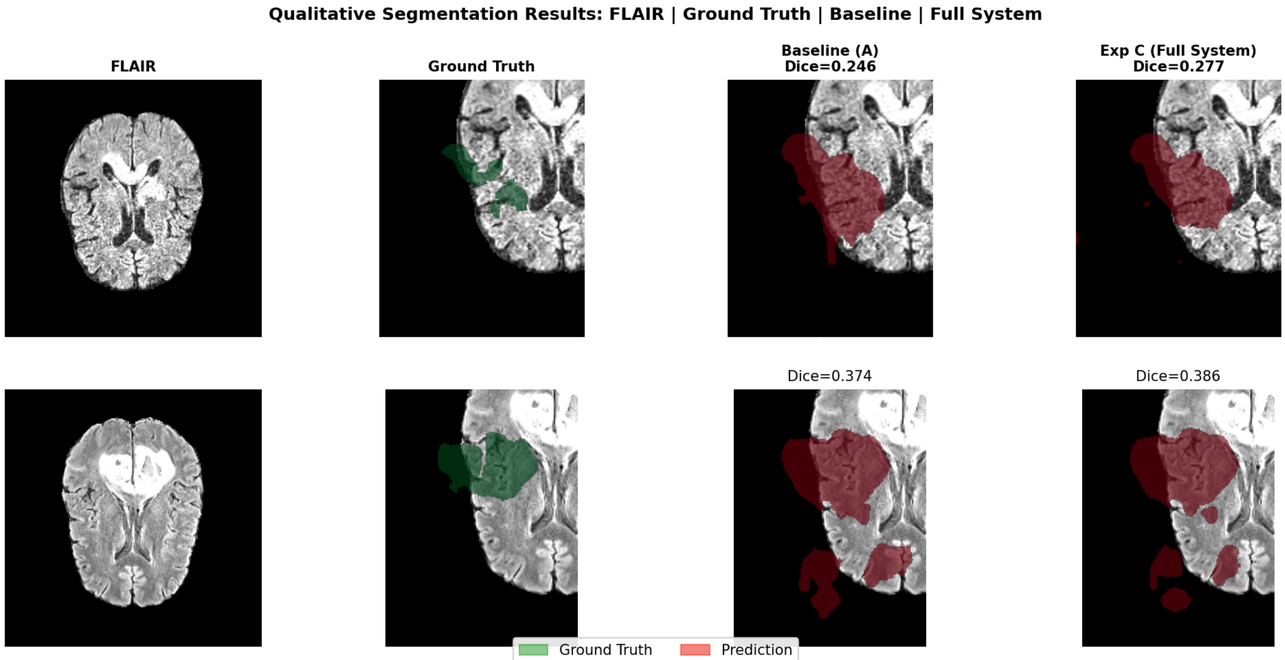

Figure 4. Representative MRI predictions across tumour size groups. Each row shows a single axial slice: FLAIR input (left), ground-truth mask overlaid in green (centre), and Experiment C prediction overlaid in red (right). Top row: small tumour, where the baseline produces no foreground while Experiment C localises the lesion. Middle row: medium tumour, where both models produce reasonable masks but Experiment C shows tighter boundary adherence. Bottom row: large tumour, where both models segment the mass well.

tal's patient mix changes over time, or if new hospitals join the federation later, the weights would need to be recalculated from scratch.

- **Untuned alpha.** We fixed α=0.5 as a reasonable starting point but never searched for the best value. The optimal blend between dataset size and small-tumour richness could be different for other datasets or federation sizes.
- **Single random seed.** All experiments used one random seed and one data partition. We cannot report how much the results would vary across different random splits.
- **Dataset scope.** All experiments are on the FeTS 2022 brain tumour dataset only. Even within this dataset, small-tumour cases were very rare: only 37 out of 338 training cases (11%), and most clients had just 0–9 small cases each. There is simply very little data to learn from for this group, which limits how much any method can improve. Results may not transfer to other tumour types or datasets where the skew pattern looks different.

6. Conclusion

We presented SCWA-LW, a federated aggregation strategy that addresses the small-tumour fairness gap by routing decoder-layer aggregation through size-composition-weighted client contributions while preserving standard FedAvg for shared encoder representations. Combined with FedProx regularisation and a strong Focal Tversky loss, the full system achieves a 10% relative improvement in small-tumour Dice and an 8.3% improvement in fair score over the FedAvg baseline, while also improving on medium and large tumour groups. A four-experiment controlled ablation confirms that stronger local supervision alone is insufficient, that FedProx regularisation *without* SCWA-LW is actively harmful to small-lesion performance (dropping small-Dice below the FedAvg baseline), and that SCWA-LW is the dominant driver: adding it over the no-SCWA control (Exp B2) delivers the full $+0.021$ small-Dice and $+0.036$ fair-score improvement. Future work should explore adaptive α scheduling that increases small-lesion bias as training progresses, variance quantification across multiple random seeds, and extension to heterogeneous federation sizes where the correlation between n_i and s_i may differ.

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